ADC Scaling v4.3 User Guide

Introduction (Ask a Question)

An analog to digital converter (ADC) converts a voltage signal as input to a digital signal in a number of bits that depends on ADC resolution. Signals that are not available in voltage form, such as currents, are converted to a voltage signal before conversion to digital data. The digital signal output of the ADC that is relative to its bit width must be scaled to a value that can be properly interpreted by the system that processes the signal. The digital output of ADC can have intended or unintended offset. An intended offset is usually generated in current measurement to allow measurement of current in positive and negative direction. The offset is usually at half of the ADC digital output which can have deviations due to tolerances in the components used in analog signal processing circuit.

The ADC Scaling IP implements to basic operations. The IP computes the ADC offset value by averaging the first 8192 ADC samples after reset release. The IP then subtracts the offset value and scales the measurement to a per unit value. The ADC scaling should be chosen such that the rated current represents 65536 output.

Summary (Ask a Question)

Core Version	This document applies to ADC scaling v4.3.
Supported Device Families	• PolarFire® SoC • PolarFire • RTG4™ • IGLOO® 2 • SmartFusion® 2
Supported Tool Flow	Requires Libero® SoC v11.8 or later releases.
Licensing	Complete encrypted RTL code is provided for the core, enabling the core to be instantiated with SmartDesign. Simulation, Synthesis, and Layout can be performed with Libero software. ADC Scaling is licensed with encrypted RTL that must be purchased separately. For more information, see ADC Scaling.

Features (Ask a Question)

ADC scaling has the following key features:

- Performs auto offset computation initially
- Involves disabling PWM and triggering the ADC by a pre-determined number of times (8192) to determine ADC offset
- Scales raw ADC data into per unit value

  $$I_{ph} = \left(adc\_result\_chX\_i - \left(\frac{ChX\_offset}{8192}\right)\right) \times \left(\frac{adc\_scale\_val\_i}{256}\right)$$

- Detects over current fault after phase current computation, which can be used to stop the motors immediately

Implementation of IP Core in Libero Design Suite (Ask a Question)

IP core must be installed to the IP Catalog of the Libero SoC software. This is done automatically through the IP Catalog update function in the Libero SoC software, or the IP core can be manually downloaded from the catalog. Once the IP core is installed in the Libero SoC software IP Catalog, the core can be configured, generated, and instantiated within the SmartDesign tool for inclusion in the Libero project list.

Device Utilization and Performance (Ask a Question)

The following table lists the device utilization used for ADC Scaling.

Table 1. ADC Scaling Utilization

Device Details		Resources		Performance (MHz)	RAMs		Math Blocks	Chip Globals
Family	Device	LUTs	DFF		LSRAM	µSRAM		
PolarFire® SoC	MPFS250T	195	138	200	0	0	1	0
PolarFire	MPF300T	195	138	200	0	0	1	0
SmartFusion® 2	M2S150	195	138	200	0	0	1	0

Important:

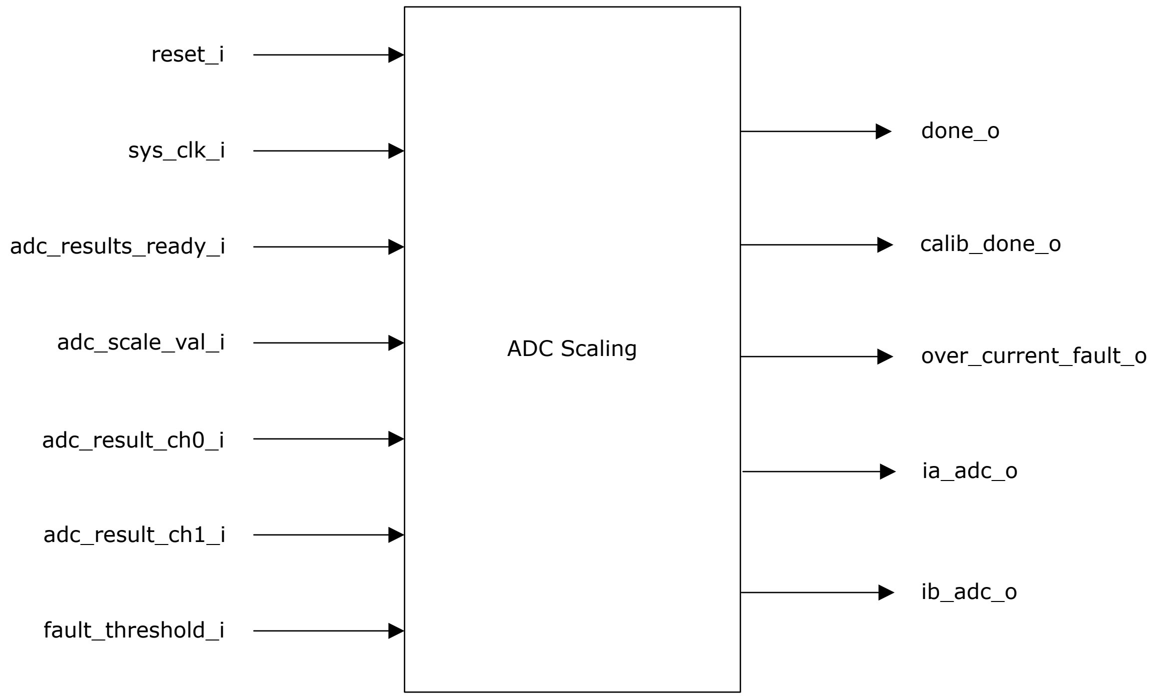
1. The data in this table is captured using typical synthesis and layout settings. CDR reference clock source was set to **Dedicated** with other configurator values unchanged.
2. Clock is constrained to 200 MHz while running the timing analysis to achieve the performance numbers.

Table of Contents

1. Functional Description (Ask a Question)

The following figure shows the block diagram of ADC scaling.

Figure 1-1. System-Level Block Diagram of ADC Scaling



The ADC scaling accumulates the first 8,192 samples of adc_result_ch0_i and adc_result_ch1_i. The calib_done_o signal is asserted after 8,192 samples are accumulated. The accumulated data is averaged and subtracted from each sample thereafter. This value is multiplied with adc_scale_val_i to give the actual ADC result. The block also asserts the over_current_fault_o when the scaled currents are more than the fault_threshold_i value.

During simulation, the block need not accumulate a large number of samples. The g_DEBUG parameter can be set to 1 to reduce the number of samples that are accumulated to 5, while setting it to 0 accumulates 8,192 samples.

2. ADC Scaling Parameters and Interface Signals (Ask a Question)

This section discusses the parameters in the ADC scaling GUI configurator and I/O signals.

2.1 Configuration of GUI Parameters (Ask a Question)

The following table lists the description of the configuration parameters used in the hardware implementation of ADC scaling. These are generic parameters and can be varied as per the requirement of the application.

Table 2-1. Configuration Parameters

Signal Name	Description
g_DEBUG	When 0, supports Synthesis When 1, supports Simulation
g_SIGNED	When 0, supports signed inputs from the ADC interface block When 1, supports unsigned inputs from the ADC interface block
g_ADC_RESULTS_WIDTH	Defines the width of the inputs from ADC measurements block
g_NO_MCYCLE_PATH	Defines the number of clock delays required before asserting the multiplier done signal

2.2 Inputs and Outputs Signals (Ask a Question)

The following table lists the input and output ports of ADC scaling.

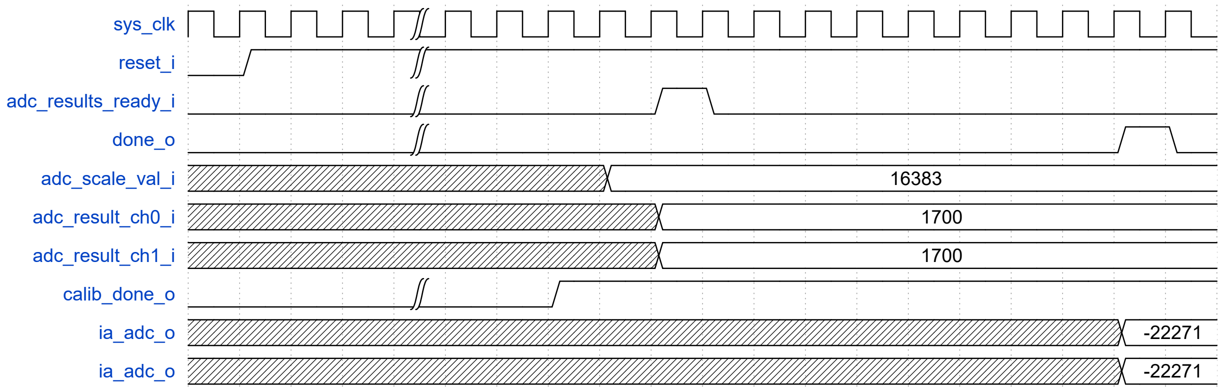
Table 2-2. Inputs and Outputs of ADC Scaling

SignalName	Direction	Description
reset_i	Input	Asynchronous active low reset signal to design
sys_clk_i	Input	System clock
adc_results_rdy_i	Input	Trigger indicating ADC inputs are available
adc_scale_val_i	Input	Scaling value for currents
adc_result_ch0_i	Input	Channel 0 result from ADC
adc_result_ch1_i	Input	Channel 1 result from ADC
fault_threshold_i	Input	Threshold value of current above which a fault is indicated
done_o	Output	Indicates completion of scaling operations – high for one clock cycle
calib_done_o	Output	A constant high signal indicates offset calibration is complete
over_current_fault_o	Output	Indicates that at least one of the currents have exceeded the fault_threshold_i input value. The signal goes back to zero (not latched) when all the currents are less than fault_threshold_i.
ia_adc_o	Output	Scaled current value for phase a (from Channel 0)
ib_adc_o	Output	Scaled current value for phase b (from Channel 1)

3. Timing Diagram (Ask a Question)

The following figure shows the timing diagram of ADC.

Figure 3-1. ADC Timing Diagram



4. Testbench (Ask a Question)

A unified test-bench is used to verify and test ADC Scaling called as user test-bench. Testbench is provided to check the functionality of the ADC Scaling IP.

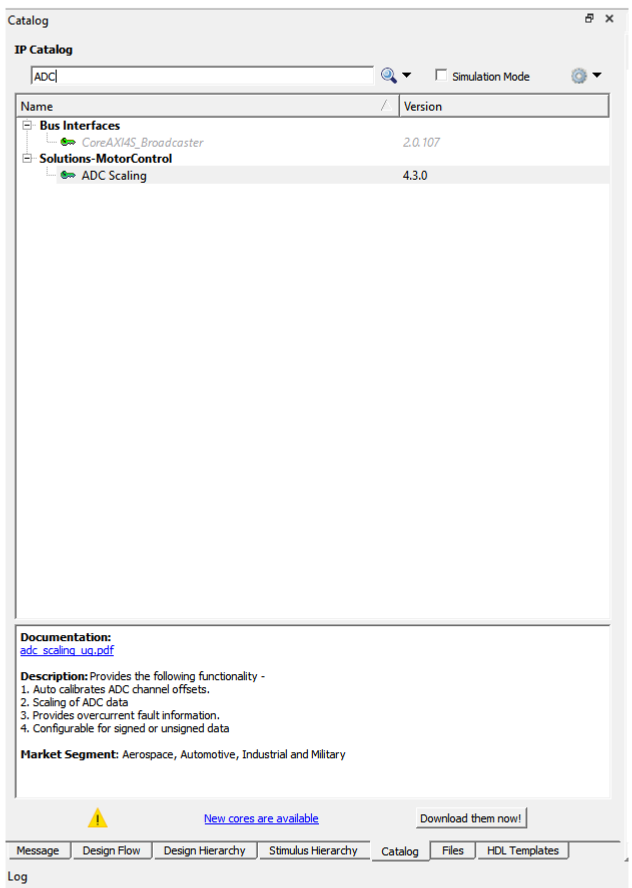
4.1 Simulation (Ask a Question)

The following steps describe how to simulate the core using the testbench:

1. Go to Libero SoC **Catalog** tab, expand **Solutions-MotorControl**, double click **ADC Scaling**, and then click **OK**. The documentation associated with the IP are listed under **Documentation**.

> **Important:** If you do not see the **Catalog** tab, navigate to **View** > **Windows** menu and click **Catalog** to make it visible.

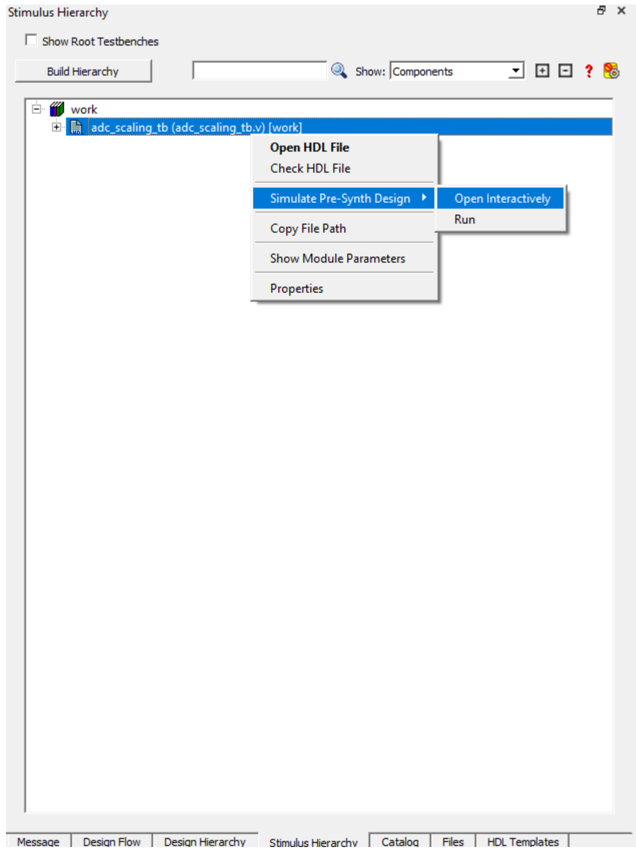
Figure 4-1. ADC Scaling IP Core in Libero® SoC Catalog



2. On the **Stimulus Hierarchy** tab, select the testbench (`adc_scaling_tb.v`), right click and then click **Simulate Pre-Synth Design** > **Open Interactively**.

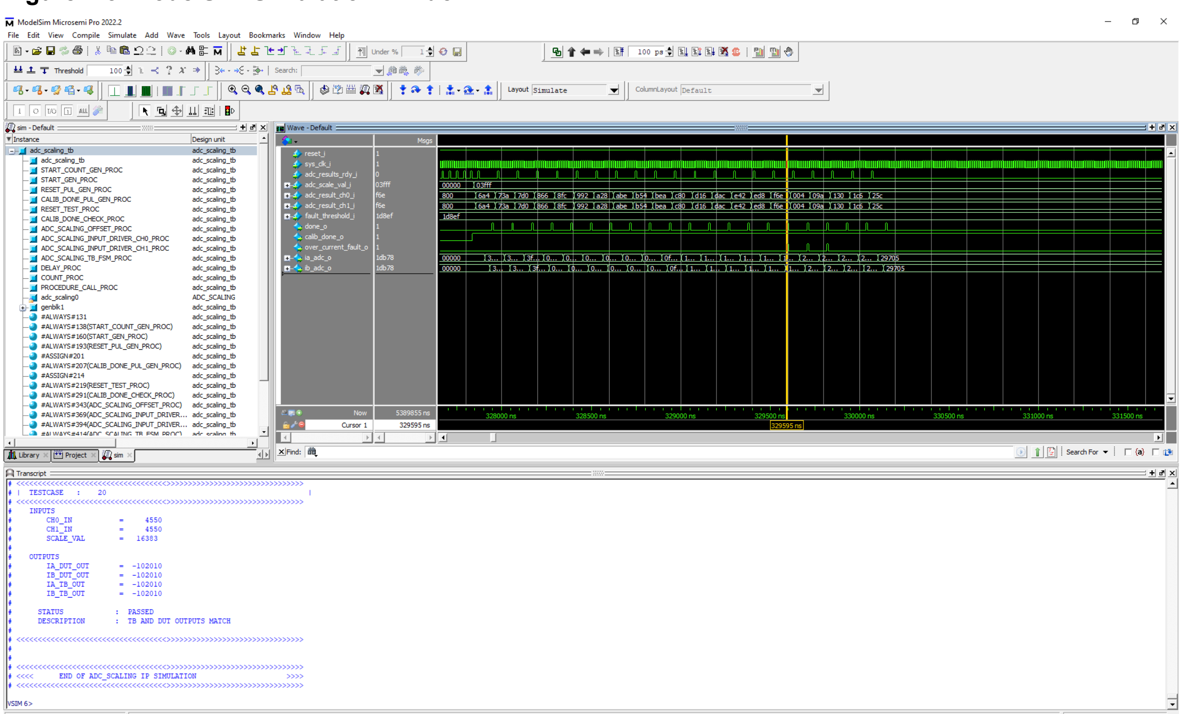
> **Important:** If you do not see the **Stimulus Hierarchy** tab, navigate to **View** > **Windows** menu and click **Stimulus Hierarchy** to make it visible.

Figure 4-2. Simulating Pre-Synthesis Design



ModelSim opens with the testbench file, as shown in the following figure.

Figure 4-3. ModelSim Simulation Window



Important: If the simulation is interrupted due to the runtime limit specified in the `.do` file, use the `run -all` command to complete the simulation.

5. Revision History (Ask a Question)

The revision history describes the changes that were implemented in the document. The changes are listed by revision, starting with the most current publication.

Table 5-1. Revision History

Revision	Date	Description
B	02/2023	The following is the list of changes in revision B of the document: • Updated the ADC scaling version from v4.2.1 to v4.3 in the document. • Updated Figure 4-1.
A	02/2023	The following is the list of changes in revision A of the document: • Migrated the document to the Microchip template. • Updated the document number to DS50003492A from 50200657. • Added 3. Timing Diagram. • Added 4. Testbench.
3.0	—	The following is a summary of changes made in this revision: • Updated the adc scaling version from v4.1 to v4.2 in the document title. • Added the g_SIGNED signal name in the Configuration Parameter table in Table 2-1. • Updated the Scales raw ADC data into phase currents formula in Introduction. • Updated the count values for Sequential elements and Combinational logic in the Resource Utilization Report of PWM Scaling table in Device Utilization and Performance.
2.0	—	This document was published in October 2016. The following was a summary of the changes in revision 2.0 of this document. • Added the IP version to the document title. • Removed g_STD_IO_WIDTH configuration parameter from 2.1. Configuration of GUI Parameters, and Device Utilization and Performance.
1.0	—	Revision 1.0 was the first publication of this document.

Microchip FPGA Support (Ask a Question)

Microchip FPGA products group backs its products with various support services, including Customer Service, Customer Technical Support Center, a website, and worldwide sales offices. Customers are suggested to visit Microchip online resources prior to contacting support as it is very likely that their queries have been already answered.

Contact Technical Support Center through the website at www.microchip.com/support. Mention the FPGA Device Part number, select appropriate case category, and upload design files while creating a technical support case.

Contact Customer Service for non-technical product support, such as product pricing, product upgrades, update information, order status, and authorization.

- From North America, call **800.262.1060**
- From the rest of the world, call **650.318.4460**
- Fax, from anywhere in the world, **650.318.8044**

Microchip Information (Ask a Question)

The Microchip Website (Ask a Question)

Microchip provides online support via our website at www.microchip.com/. This website is used to make files and information easily available to customers. Some of the content available includes:

- **Product Support** – Data sheets and errata, application notes and sample programs, design resources, user's guides and hardware support documents, latest software releases and archived software
- **General Technical Support** – Frequently Asked Questions (FAQs), technical support requests, online discussion groups, Microchip design partner program member listing
- **Business of Microchip** – Product selector and ordering guides, latest Microchip press releases, listing of seminars and events, listings of Microchip sales offices, distributors and factory representatives

Product Change Notification Service (Ask a Question)

Microchip's product change notification service helps keep customers current on Microchip products. Subscribers will receive email notification whenever there are changes, updates, revisions or errata related to a specified product family or development tool of interest.

To register, go to www.microchip.com/pcn and follow the registration instructions.

Customer Support (Ask a Question)

Users of Microchip products can receive assistance through several channels:

- Distributor or Representative
- Local Sales Office
- Embedded Solutions Engineer (ESE)
- Technical Support

Customers should contact their distributor, representative or ESE for support. Local sales offices are also available to help customers. A listing of sales offices and locations is included in this document.

Technical support is available through the website at: www.microchip.com/support

Microchip Devices Code Protection Feature (Ask a Question)

Note the following details of the code protection feature on Microchip products:

- Microchip products meet the specifications contained in their particular Microchip Data Sheet.
- Microchip believes that its family of products is secure when used in the intended manner, within operating specifications, and under normal conditions.
- Microchip values and aggressively protects its intellectual property rights. Attempts to breach the code protection features of Microchip product is strictly prohibited and may violate the Digital Millennium Copyright Act.
- Neither Microchip nor any other semiconductor manufacturer can guarantee the security of its code. Code protection does not mean that we are guaranteeing the product is “unbreakable”. Code protection is constantly evolving. Microchip is committed to continuously improving the code protection features of our products.

Legal Notice (Ask a Question)

Trademarks (Ask a Question)

ISBN: 978-1-6683-2065-5

Quality Management System (Ask a Question)

For information regarding Microchip's Quality Management Systems, please visit www.microchip.com/quality.

Worldwide Sales and Service

AMERICAS	ASIA/PACIFIC	ASIA/PACIFIC	EUROPE
Corporate Office 2355 West Chandler Blvd. Chandler, AZ 85224-6199 Tel: 480-792-7200 Fax: 480-792-7277 Technical Support: www.microchip.com/support Web Address: www.microchip.com **Atlanta** Duluth, GA Tel: 678-957-9614 Fax: 678-957-1455 **Austin, TX** Tel: 512-257-3370 **Boston** Westborough, MA Tel: 774-760-0087 Fax: 774-760-0088 **Chicago** Itasca, IL Tel: 630-285-0071 Fax: 630-285-0075 **Dallas** Addison, TX Tel: 972-818-7423 Fax: 972-818-2924 **Detroit** Novi, MI Tel: 248-848-4000 **Houston, TX** Tel: 281-894-5983 **Indianapolis** Noblesville, IN Tel: 317-773-8323 Fax: 317-773-5453 Tel: 317-536-2380 **Los Angeles** Mission Viejo, CA Tel: 949-462-9523 Fax: 949-462-9608 Tel: 951-273-7800 **Raleigh, NC** Tel: 919-844-7510 **New York, NY** Tel: 631-435-6000 **San Jose, CA** Tel: 408-735-9110 Tel: 408-436-4270 **Canada - Toronto** Tel: 905-695-1980 Fax: 905-695-2078	**Australia - Sydney** Tel: 61-2-9868-6733 **China - Beijing** Tel: 86-10-8569-7000 **China - Chengdu** Tel: 86-28-8665-5511 **China - Chongqing** Tel: 86-23-8980-9588 **China - Dongguan** Tel: 86-769-8702-9880 **China - Guangzhou** Tel: 86-20-8755-8029 **China - Hangzhou** Tel: 86-571-8792-8115 **China - Hong Kong SAR** Tel: 852-2943-5100 **China - Nanjing** Tel: 86-25-8473-2460 **China - Qingdao** Tel: 86-532-8502-7355 **China - Shanghai** Tel: 86-21-3326-8000 **China - Shenyang** Tel: 86-24-2334-2829 **China - Shenzhen** Tel: 86-755-8864-2200 **China - Suzhou** Tel: 86-186-6233-1526 **China - Wuhan** Tel: 86-27-5980-5300 **China - Xian** Tel: 86-29-8833-7252 **China - Xiamen** Tel: 86-592-2388138 **China - Zhuhai** Tel: 86-756-3210040	**India - Bangalore** Tel: 91-80-3090-4444 **India - New Delhi** Tel: 91-11-4160-8631 **India - Pune** Tel: 91-20-4121-0141 **Japan - Osaka** Tel: 81-6-6152-7160 **Japan - Tokyo** Tel: 81-3-6880- 3770 **Korea - Daegu** Tel: 82-53-744-4301 **Korea - Seoul** Tel: 82-2-554-7200 **Malaysia - Kuala Lumpur** Tel: 60-3-7651-7906 **Malaysia - Penang** Tel: 60-4-227-8870 **Philippines - Manila** Tel: 63-2-634-9065 **Singapore** Tel: 65-6334-8870 **Taiwan - Hsin Chu** Tel: 886-3-577-8366 **Taiwan - Kaohsiung** Tel: 886-7-213-7830 **Taiwan - Taipei** Tel: 886-2-2508-8600 **Thailand - Bangkok** Tel: 66-2-694-1351 **Vietnam - Ho Chi Minh** Tel: 84-28-5448-2100	**Austria - Wels** Tel: 43-7242-2244-39 Fax: 43-7242-2244-393 **Denmark - Copenhagen** Tel: 45-4485-5910 Fax: 45-4485-2829 **Finland - Espoo** Tel: 358-9-4520-820 **France - Paris** Tel: 33-1-69-53-63-20 Fax: 33-1-69-30-90-79 **Germany - Garching** Tel: 49-8931-9700 **Germany - Haan** Tel: 49-2129-3766400 **Germany - Heilbronn** Tel: 49-7131-72400 **Germany - Karlsruhe** Tel: 49-721-625370 **Germany - Munich** Tel: 49-89-627-144-0 Fax: 49-89-627-144-44 **Germany - Rosenheim** Tel: 49-8031-354-560 **Israel - Ra'anana** Tel: 972-9-744-7705 **Italy - Milan** Tel: 39-0331-742611 Fax: 39-0331-466781 **Italy - Padova** Tel: 39-049-7625286 **Netherlands - Drunen** Tel: 31-416-690399 Fax: 31-416-690340 **Norway - Trondheim** Tel: 47-72884388 **Poland - Warsaw** Tel: 48-22-3325737 **Romania - Bucharest** Tel: 40-21-407-87-50 **Spain - Madrid** Tel: 34-91-708-08-90 Fax: 34-91-708-08-91 **Sweden - Gothenberg** Tel: 46-31-704-60-40 **Sweden - Stockholm** Tel: 46-8-5090-4654 **UK - Wokingham** Tel: 44-118-921-5800 Fax: 44-118-921-5820